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A predict-and-prescribe framework for dynamic course scheduling toward strategic university scaling☆

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ABSTRACT

Universities play vital roles in educating current and future generations. Universities that intend to survive in competitive academic markets where there are evolving enrollment trends must responsibly manage resources. Planning processes for instructional spaces can benefit from optimization and lead to more effective resource utilization, yet the use of trends in major and course demands to inform long-term planning remains largely unexplored. We propose a novel mathematical optimization framework that appears to be the first to use trend predictions to guide long-term dynamic course scheduling decisions and effective resource allocation. We begin with a baseline formulation that assigns course sections to time patterns and classroom spaces while considering instructor preferences, as well as constraints related to conflicts and capacity. We then extend this to a dynamic formulation that addresses bottleneck courses while prioritizing classroom utilization and honoring instructor preferences. Our dynamic formulation optimizes instructional space allocation for an academic period over the entire university, and solving for sequential independent academic periods reveals valuable insights into the efficiency of physical resource utilization on the horizon. Our framework is flexible, accommodating the objectives of faculty, schedulers, and administrators through a hierarchical multi-objective approach that integrates these diverse priorities. Our formulation addresses 76% and 81% of bottleneck sections for back-to-back semesters, with a substantial number of unutilized locations that could potentially be repurposed to accommodate the remaining bottleneck sections or other purposes. Through extensive experiments with varying university enrollment scenarios, we examine the resulting tradeoffs among objectives and highlight a variety of implications.

1. Introduction

The past quarter-century has brought unprecedented challenges to universities such as economic pressures, increased competition, and technological advancements, causing planning operations to constantly adapt [1]. The impending enrollment cliff of 2026 is a projected demographic downturn expected to reduce college enrollment [2]. At the same time, student preferences are shifting toward more flexible, affordable, and career-oriented programs, driven by demographic shifts, economic circumstances, and university body expansion [3–5]. Such dynamics call for new approaches to better utilize university instructional spaces while freeing resources for strategic university priorities. The benefits of such practices extend beyond fiscal responsibility to increasing student satisfaction, campus sustainability, and securing competitive advantage [6,7].

University timetabling, or *course scheduling*, involves the challenging process of allocating courses to locations and times while ensuring a diverse range of university-specific constraints. Universities use a variety of approaches for scheduling including ad hoc manual methods and commercial scheduling software, together with adjustments for adapting year-over-year course schedules. However, the availability of faculty can intensify scheduling demands, which causes manual modifications (even with commercial software) to become time consuming and increasingly challenging to manage, often yielding schedules that are far from optimal [8–10]. Extended planning of instructional spaces for the entire university, especially over a time horizon of multiple years and with fluctuating enrollment, course and major trends, remains an elusive goal.

Mathematical optimization has been used to assist universities in identifying optimal or near-optimal solutions to scheduling challenges,

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allowing for rapid and efficient adjustments to schedules via a wide range of constraints and objective functions. Optimizing for any variety of utilization-related objectives can already reveal the institutional capacity of available spaces at intermediate levels, such as department or school-level planning. While course timetabling has been shown effective at sub-university levels [11], these inefficiencies compound and as such, timetabling studies rarely address the entire university, and even more so with changing enrollment, course and major trends over time. Thus at the university-wide level, particularly for addressing shifting enrollment trends and long-term planning, mathematical optimization remains unexplored. We aim to address a key question that universities face regarding sustainable resource allocation under fluctuating campus enrollments: *How can instructional space capacity at universities be efficiently planned over time by leveraging predicted major and course trends?*

We propose a predict-and-prescribe framework that enables universities to evaluate expansion and contraction scenarios over multiple years revealing strategic insights for future decision making, for instance hiring new faculty, optimizing available instructional spaces, or adding additional course sections, all while honoring instructor preferences and space limitations. We first forecast major distributions and course demands for future years. Yet, overlapping instructor preferences while maintaining responsible classroom densities and evolving student preferences can lead to conflicts that we refer to as *bottleneck sections*. We dynamically address bottleneck sections by allowing for relaxing reduced capacity constraints and splitting larger sections into available locations and time patterns. We introduce a multiobjective mixed-integer optimization formulation that hierarchically orders multiple efficiency-based objective functions, while ensuring timetable-related constraints.

We make several novel contributions to the literature. (1) We introduce a framework for universities to conduct strategic instructional space planning by estimating enrollment in light of fluctuations, major and course trends, and driven by a course scheduling engine powered through mathematical optimization, marking the first study in the course scheduling literature to combine both prediction and prescription. (2) We extend our baseline formulation to efficiently address bottleneck sections by dynamic splitting and fitting into smaller instructional spaces and relaxing reduced capacity requirements. (3) We are among the first to address university-wide timetabling, while also accounting for overlapping academic periods within a single semester. (4) We conduct a variety of sensitivity analyses under fluctuating enrollments and major and course trends, namely (4a) optimizing university-wide course schedules over many time periods to understand the effect of the absence of popular buildings, (4b) introducing tradeoffs for decision makers to assess between adding more course sections or increasing capacities and retaining bottleneck sections, (4c) conducting a retrospective analysis to evaluate the performance of our framework on utilizing available instructional space, and (4d) performing strategic analysis for additional course sections at the departmental level and evaluating location capacities for expansion.

The remainder of our study is structured as follows. Section 2 covers related background material from the literature. Section 3 introduces our mathematical modeling, including the baseline and dynamic formulations, as well as our multiobjective framework. Section 4 presents datasets, predictive approach, and extensive computational experiments. Section 5 reveals computational results and insights, while Section 6 concludes our study.

2. Literature review

Timetabling problems originate in diverse forms and contexts such as education [12], health [13], transportation [14], and sports [15]. University course timetabling (CTT) stands as a notable area within educational timetabling, alongside others such as conference timetabling [16], examination timetabling [17], and high school timetabling [18], and consists of three main variants: (1) *Post-Enrollment Course Timetabling* (PE-CTT) [19,20] involves scheduling events, periods, and rooms after students enroll in courses, aiming to minimize conflicts between overlapping events, while accounting for room features and capacities, precedence and unavailability of events, and penalizes inefficient scheduling, such as having students attend multiple successive events; (2) *Curriculum-Based Course Timetabling* (CB-CTT) focuses on predefined curricula and schedules courses attended by common groups of students, with conflicts determined by curriculum overlaps, while ensuring constraints such as teacher availability, room occupancy, and aims to optimize room capacity usage and lecture spread across days; (3) *ITC-2019*[1] combines CTT with student sectioning, focusing on assigning classes to periods and rooms based on course enrollments, to effectively manage immediate and detailed aspects of course scheduling, yet it does not address long-term decisions such as long-term strategic resource allocation. Our framework aligns most closely with CB-CTT, particularly in managing course conflicts for students. However, we only common course groupings based on student majors and class years, diverging from the predefined curricula of CB-CTT.

A wide variety of methods have been explored to address complex academic scheduling problems. The importance to delivery mode, meeting pattern, and room assignment preference has increased the complexity, often making it computationally challenging to solve instances in reasonable times. Petrovic and Burke (2004) discuss several solution methods for these challenges in academic scheduling, such as graph coloring, cluster and meta-heuristic methods, constraint based, multicriteria, and case-based reasoning approaches [22]. Among these, one of the most well-used techniques is integer optimization, as it effectively addresses the detailed and varied aspects of academic scheduling, ensuring accommodation for a diverse set of requirements and preferences. Integer optimization effectively complements multicriteria approaches, also known as hierarchical optimization, which are often used to address real-world timetabling problems, as they allow for constraints to be quantitatively measured and optimized [22]. A hierarchical optimization approach, where components of the objective function are assigned different hierarchical ranks, is proposed in the studies by van den Broek et al. (2009) and Navabi-Shirazi et al. (2022) [23,24]. The benefits of trading off multiple objectives are well-explored in both papers. Lach and Lübbecke (2012) propose decomposing the CB-CTT problem into two stages via integer optimization, that is, to first assign meeting patterns p to lectures s, and secondly, assign these (s, p) pairs to locations [25]. Their proposed approach both proves optimality and outperforms the strongest known heuristic-based methods.

Studies involving predictive analytics in higher education primarily focus on forecasting student performance and identifying student behaviors [26]. Alyahyan and Düştegör (2020) provide comprehensive guidelines for non-technical users for predicting student success [26]. They present a six-stage framework designed to simplify the process, emphasizing its adaptability for undergraduates across disciplines and the potential to include additional data, such as faculty information, to discover new success factors. Aydoğdu (2020) investigates the prediction of student performance in online learning environments using an artificial neural network model [28]. Teng et al. (2023) aim to forecast student GPAs and determine whether students intend to complete courses [29]. They further suggest that data-driven decision-making could include analyzing enrollment patterns for specific departments, yet do not carry out such analyses. To the best of our knowledge, there is no study that predicts enrollment and course pattern trends across different majors within a university. We predict enrollment based on major and course trends, using these signals for optimizing future course planning.

While strategic decisions have long-term effects on university timetables, the literature has largely overlooked strategic problems [36]. An

[1] It is proposed for the ITC-2019 competition [21].

Table 1
Academic studies involving strategic decisions in the timetabling literature.

Reference	Model Side					Managerial Side		
	Method	Hard Constraints	Objective(s)	Course Splitting	Data	Room Planning	Enrollment Scaling	Prediction
Fizzano and Swanson (2000)	Heuristics	Location-Time Conflict	Minimum location	✗	Real Data (University wide)	✓	✗	✗
Beyrouthy et al. (2007)	Multiple Scenario Method Heuristics	Capacity	Room-Size Profiles	✗	Real Data (Sub-university wide)	✓	✗	✗
Beyrouthy et al. (2007)	Simulated Annealing	Capacity Location-Time Conflict	Utilization Course Conflicts Travel Distance Section Size	✓	Real Data (Sub-university wide)	✓	✗	✗
Beyrouthy et al. (2008)	Integer Optimization	Capacity Location-Time Conflict Group Size Penalty	Maximum Utilization	✓	Real Data (Sub-university wide)	✓	✗	✗
Beyrouthy et al. (2009)	Integer Optimization Local Search	Capacity Location-Time Conflict	Utilization Location Preference Course Conflicts	✗	Real Data (Sub-university wide)	✓	✗	✗
Beyrouthy et al. (2010)	Integer Optimization	Capacity Location-Time Conflict Location Penalty Group Size Penalty Space Type Mix Penalty Group Size Penalty Number of Splits Course Mode Conflicts	Maximum Utilization	✓	Real Data (Sub-university wide)	✓	✗	✗
Lindahl et al. (2018)	Mixed Integer Linear Optimization	Capacity Location-Time Conflict Course Conflicts Instructor Conflicts	Minimum Location Minimum teaching periods Quality of Timetable (Minimum Working Days + Curriculum Compactness)	✗	Real Data (Sub-university wide)	✓	✗	✗
This work	Mixed Integer Linear Optimization	Capacity Location-Time Conflict Course Conflicts Instructor Conflicts Course Mode Conflicts	Split+Bottleneck Section Maximize Soft Preference Minimum Location Maximum Utilization	✓	Real Data (University wide)	✓	✓	✓

early study by Fizzano and Swanson (2000) addresses the timetabling problem from a room planning perspective, and while their focus is operational, the minimization of room usage can influence strategic decisions regarding future resource management and infrastructure development [30,31]. Using local search and simulated annealing, Beyrouthy et al. (2007) address the teaching space allocation problem by implementing dynamic splitting, which allows for courses to be split into multiple sections or also rejoining [32]. The study enforces constraints on capacity and room-time slots and also included randomly generated pairs of events that should not conflict as part of a multiobjective function, given that the teaching space allocation problem is constrained by timetabling needs [34]. They conclude that different penalties and constraints have a significant impact on the utilization of locations. Beyrouthy et al. (2008) focus on the interaction of splitting with management and planning to achieve higher utilization in the use of spaces [33]. They present an integer optimization formulation for assigning students to room-time slot pairs, allowing the splitting of classes into smaller groups; no course conflict constraints or timetabling conflicts such as instructor conflicts were present in the study. Specifically, the study aims to identify the thresholds at which the computational complexity of the space allocation problem increases significantly. Beyrouthy et al. (2010) further extend their work by presenting an integer optimization formulation to decide the optimal policy on types of spaces, that is, various types of rooms such as lecture halls and tutorial rooms [35]. This paper appears to be the first integer optimization model that has incorporated both the space planning and timetabling aspects. They do not, however, include timetabling constraints such as instructors that have a significant effect on the quality of timetabling, nor do they include predicting trends and timetabling over a time horizon. de la Torre et al. (2016) propose a mixed-integer model for long-term staff composition planning, aiming to optimize academic staff distribution across various departments according to field expertise and specific staff categories [37]. Lindahl et al. (2018) present a strategic view of university timetabling via multiobjective mixed integer optimization and discuss long-term decisions on room planning and teaching periods [36]. They explore trade-offs between objectives such as room availability, teaching periods, and quality, generating multiple Pareto optimal solutions to guide decision makers in space allocation. However, their study does not consider critical factors such as faculty availability or changing student demand over time, both of which are integral to long-term decisions. Table 1 details a number of features of academic scheduling studies that include planning and strategic decisions. The papers were selected based on their focus on long-term strategies and their relevance to the objectives of our study.

While recent studies have addressed teaching space allocation problems, they have yet to fully incorporate timetabling conflicts or address long-term capacity planning at the university-wide level, particularly in response to fluctuating enrollments and evolving trends in majors and courses. To the best of our knowledge, our study presents the first optimization-based framework to address all of these challenges.

Table 2
Sets and parameters used in our mathematical modeling.

Symbol	Description
Primary Sets	
$\mathcal{C}$	Set of all courses, indexed by c
$\mathcal{S}$	Set of all course sections, indexed by s
$\mathcal{L}$	Set of all instructional locations (spaces), indexed by ℓ
$\mathcal{P}$	Set of all patterns, indexed by p
$\mathcal{I}$	Set of all instructors, indexed by i
$\mathcal{D}$	Set of days in a week: {M, T, W, R, F}, indexed by d
$\mathcal{H}$	Time slots for a day, indexed by h
$\mathcal{DH}$	Set of time blocks in a week, indexed by (d,h)
$\mathcal{F}$	Set of all instructional formats, indexed by f
$\mathcal{K}$	Set of (possibly overlapping) academic periods, indexed by k
$\hat{\mathcal{C}}$	Set of triplets of common courses, indexed by g
$\mathcal{N}$	Set of university scaling enrollment scenarios, indexed by n, where n represent changes in numbers of students enrolled
Subsets	
$\mathcal{S}_c$	Set of all course sections that belong to course c
$\mathcal{S}_\ell$	Set of all course sections that can be assigned to room location ℓ
$\mathcal{S}_i$	Set of all course sections that are taught by instructor i
$\mathcal{S}_k$	Set of all course sections that are offered during term k
$\mathcal{S}_{(g,k)}$	Set of common course sections for the group of common courses $g \in \hat{\mathcal{C}}$ during period k
$\mathcal{S}_f$	Set of course sections that are taught in instructional format f
$\mathcal{L}_s$	Set of all available locations for course section s
$\mathcal{L}_T$	Set of all teaching spaces
$\mathcal{L}_B$	Set of all lab spaces
$\mathcal{DH}_i$	Set of time blocks in a week to which instructor i can be assigned
$\mathcal{P}_s$	Set of acceptable patterns for course section s
$\mathcal{P}_s^{(d,h)}$	Set of all conflicting patterns with time block (d,h), for course section s
Parameters	
c_ℓ	Absolute (maximum) physical capacity of location $\ell \in \mathcal{L}$
λ_ℓ	Percent of absolute physical capacity allowed for room location ℓ
$\hat{r}_s^n$	Estimate of the number of registered students for course section $s \in \mathcal{S}$, when enrollments are scaled according to scenario $n \in \mathcal{N}$

3. Mathematical modeling and formulations

We introduce a baseline mathematical formulation and further extend it to allow for dynamically splitting course sections and relaxing space capacity restrictions to address bottleneck course sections. The baseline model assigns courses to classrooms and patterns while minimizing the number of bottleneck sections. The dynamic splitting and capacity expansion formulation addresses bottleneck sections while prioritizing instructor preferences and effective use of physical resources. The mathematical sets and parameters for the remainder of the paper are contained in Table 2.

3.1. Mathematical preliminaries

Academic calendars can be divided into one or more periods, such as fall, spring, and summer semesters. While many institutions have distinct academic periods over which courses are scheduled, at other institutions these periods may vary in length or occur simultaneously (*overlapping academic periods*). *Courses* are typically a single class on a particular subject, such as Calculus I or Social Psychology. *Course sections* are distinct offerings of a course, which take on a particular *instructional format* such as lecture, laboratory, or discussion. In the process of scheduling, there may be course sections that cannot be assigned to an available location and time pattern due to reasons such as jointly overlapping instructor availability for limited classroom space, reduced location capacity, or enrollment size that exceeds the allocated capacity of classrooms. We refer to these as *bottleneck sections*, which are common in all universities when unresolvable scheduling conflicts arise.

Our proposed optimization-based framework can address many of these conflicts, with any that remain resolvable via manual processes such as communication and negotiation, a reasonable yet cumbersome task faced by university administrators. One way we propose to handle bottleneck sections is by splitting into subsections and assigning into smaller available locations and time patterns. We refer to these subsections as *splits*, which subsequently create *additional course sections* that must be accommodated on campus. Moreover, certain course sections are restricted to specific buildings and instructional spaces due to specialized laboratory requirements. When these buildings become unavailable, whether due to planned or unforeseen factors, it can create significant scheduling challenges. These sections, strained by space limitations and greater student registrations, are what we refer to as *near-bottleneck sections*.

There are *instructional spaces* where course sections can be scheduled according to a specific pattern. These spaces are categorized as *teaching and lab spaces* with associated attributes including absolute and relative capacity, the latter of which depends upon the proportion up to 100% that is permissible to fill. For scheduling purposes, each day is divided into a set of *time blocks, or slots*, of fixed time durations. For example, hour-long block durations yield time blocks such as Monday 08:00 AM to 09:00 AM, Monday 09:00 AM to 10:00 AM, and so on. The combination of days with time slots is defined as a *meeting pattern* in which courses are to be scheduled. An example of a meeting pattern could be "TF2PM350PM" which is to be interpreted as a course being offered on Tuesday and Friday between 2:00 PM and 3:50 PM. This meeting pattern *conflicts* with four time blocks: Tuesday 2:00 PM to 3:00 PM, Tuesday 3:00 PM to 4:00 PM, Friday 2:00 PM to 3:00 PM and Friday 3:00 PM to 4:00 PM.

We next propose our mathematical optimization formulations to address challenges related to scaling enrollment and shifting trends in Sections 3.2 and 3.3. We then introduce our multiobjective optimization approach that considers multiple objectives hierarchically in Section 3.4.

3.2. Baseline formulation

Let binary variable $x_s^{p\ell}$ equal 1 if course section $s \in \mathcal{S}$ is scheduled for location $\ell \in \mathcal{L}_s$ according to pattern $p \in \mathcal{P}_s$, and 0 otherwise. The

Table 3
Additional notation for the DSCE formulation.

Symbol	Description
Parameters	
t_s	Binary parameter indicating whether course section $s \in S$ is splittable and the capacity of the assigned location can be expanded
L_s	Lower bound on the number of students for split $s \in S$
U_s	Upper bound on the number of students for split $s \in S$
μ_ℓ	Maximum allowed capacity expansion for location ℓ
$v_s^{p\ell}$	Preference value of assigning course section $s \in S$ to location ℓ according to pattern p

baseline formulation under enrollment scenario n is:

$$\text{maximize} \quad \sum_{s\in S}\sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \tag{1a}$$

$$\text{subject to} \quad \sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \le 1, \quad \forall\, s\in S, \tag{1b}$$

$$\sum_{p\in\mathcal{P}_s} \hat{r}_s^n x_s^{p\ell} \le \lambda_\ell c_\ell, \quad \forall\, s\in S,\ \ell\in\mathcal{L}_s, \tag{1c}$$

$$\sum_{s\in\{S_\ell\cap S_k\}}\sum_{p\in\mathcal{P}_s^{(d,h)}} x_s^{p\ell} \le 1, \quad \forall\, \ell\in\mathcal{L},\ k\in\mathcal{K},\ (d,h)\in\mathcal{DH}, \tag{1d}$$

$$\sum_{s\in\{S_i\cap S_k\}}\sum_{p\in\mathcal{P}_s^{(d,h)}}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \le 1, \quad \forall\, i\in\mathcal{I},\ k\in\mathcal{K},\ (d,h)\in\mathcal{DH}_i, \tag{1e}$$

$$\sum_{s\in\{S_c\cap S_k\}}\sum_{p\in\mathcal{P}_s^{(d,h)}}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \le \alpha, \quad \forall\, c\in C,\ k\in\mathcal{K},(d,h)\in\mathcal{DH}, \tag{1f}$$

$$\sum_{s\in\hat{S}_{(g,k)}}\sum_{p\in\mathcal{P}_s^{(d,h)}}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \le \beta, \quad \forall\, g\in\hat{C},\ k\in\mathcal{K},\ (d,h)\in\mathcal{DH}, \tag{1g}$$

$$\sum_{s\in\{S_c\cap S_k\cap S_f\}}\sum_{p\in\mathcal{P}_s^{(d,h)}}\sum_{\ell\in\mathcal{L}_s} x_s^{p\ell} \le \gamma, \quad \forall\, c\in C,\ k\in\mathcal{K},\ f\in\mathcal{F},\ (d,h)\in\mathcal{DH}, \tag{1h}$$

$$x_s^{p\ell} \in \{0,1\}, \quad \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s. \tag{1i}$$

Objective function (1a) maximizes the number of course sections assigned to a meeting pattern and location, alternatively minimizing the number of bottleneck sections. Constraint set (1b) ensures that every course section is assigned at most once. Constraint set (1c) ensures that adjusted location capacity is honored. Constraint sets (1d) and (1e) prevent location and instructor conflicts by assigning at most one course section during a time block for a location and an instructor for each overlapping term. Constraint set (1f) upper bounds the number of course sections of each course that may be assigned during overlapping time blocks for each overlapping term. Constraint set (1g) upper bounds the number of course sections of each common course triplet during overlapping time blocks for each overlapping term. Constraint set (1h) upper bounds the number of course sections of each course in the same instructional format that may be assigned during overlapping time blocks for each overlapping term. The values of α, β and γ are further elaborated in the experimental settings. Variable domains are expressed in (1i).

A solution to the baseline formulation maximizes the number of scheduled course sections while satisfying stated constraints. The difference between the number of courses $|S|$ to schedule and the optimal objective function value of (1a) is the number of bottleneck sections, and when positive, can be further addressed with our dynamic splitting and capacity expansion formulation, which we next introduce.

3.3. Dynamic splitting and capacity expansion (DSCE) formulation

The dynamic splitting and capacity expansion (DSCE) formulation addresses the issue of bottleneck sections by splitting and expanding capacity for specific course sections that would otherwise become bottlenecks. Additional notation for formulating the DSCE model, and specifying which course sections can be split and the capacity of the assigned location can be expanded, is given in Table 3.

Let binary variable y_ℓ equal 1 if location $\ell \in \mathcal{L}$ is utilized, and 0 otherwise. Similarly, auxiliary binary variable w_s equals 1 if course section $s \in S$ is assigned, and 0 otherwise. In addition, binary variable $z_s^{p\ell}$ equals 1 if course section $s \in S$ is split to location $\ell \in \mathcal{L}_s$ and pattern $p \in \mathcal{P}_s$, and 0 otherwise. The variable $q_s^{p\ell}$ denotes the proportion of students of course section $s \in S$ split to location $\ell \in \mathcal{L}_s$, and pattern $p \in \mathcal{P}_s$. Lastly, θ_s^ℓ represents the extent of capacity relaxation for course section $s \in S$ in location $\ell \in \mathcal{L}_s$.

The system of constraints needed to express the DSCE formulation for enrollment scenario n are given in the following mathematical formulation.

$$\sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s}(x_s^{p\ell}+q_s^{p\ell}) = w_s, \quad \forall\, s\in S, \tag{2a}$$

$$\hat{r}_s^n(x_s^{p\ell}+q_s^{p\ell}) \le (\lambda_\ell+\theta_s^\ell)c_\ell, \quad \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s, \tag{2b}$$

$$\lambda_\ell+\theta_s^\ell \le \mu_\ell, \quad \forall\, s\in S,\ \ell\in\mathcal{L}_s, \tag{2c}$$

$$\hat{r}_s^n q_s^{p\ell} \le U_s z_s^{p\ell}, \quad \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s, \tag{2d}$$

$$\hat{r}_s^n q_s^{p\ell} \ge L_s z_s^{p\ell}, \quad \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s, \tag{2e}$$

$$z_s^{p\ell} \le t_s, \quad \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s, \tag{2f}$$

$$\theta_s^\ell \le t_s, \quad \forall\, s\in S,\ \ell\in\mathcal{L}_s, \tag{2g}$$

$$\sum_{s\in\{S_\ell\cap S_k\}}\sum_{p\in\mathcal{P}_s^{(d,h)}}(x_s^{p\ell}+z_s^{p\ell}) \le y_\ell, \quad \forall\, \ell\in\mathcal{L},\ k\in\mathcal{K},\ (d,h)\in\mathcal{DH}, \tag{2h}$$

$$\sum_{s\in\{S_i\cap S_k\}}\sum_{p\in\mathcal{P}_s^{(d,h)}}\sum_{\ell\in\mathcal{L}_s}(x_s^{p\ell}+z_s^{p\ell}) \le 1, \quad \forall\, i\in\mathcal{I},\ k\in\mathcal{K},\ (d,h)\in\mathcal{DH}_i, \tag{2i}$$

expressions (1f), (1g), (1h), (1i),

$$w_s \in \{0,1\}, \quad \forall\, s\in S, \tag{2j}$$

$$y_\ell \in \{0,1\}, \quad \forall\, \ell\in\mathcal{L}, \tag{2k}$$

$$z_s^{p\ell} \in \{0,1\}, \quad q_s^{p\ell} \in \mathbb{R}_+, \ \forall\, s\in S,\ p\in\mathcal{P}_s,\ \ell\in\mathcal{L}_s, \tag{2l}$$

$$\theta_s^\ell \in \mathbb{R}_+, \quad \forall\, s\in S,\ \ell\in\mathcal{L}_s. \tag{2m}$$

Constraint set (2a) assigns course sections to meeting patterns and locations while determining the status of sections: whether they are assigned in standard fashion, proportioned (split) into smaller sections, or relegated to bottleneck sections. If a section is split, the equation also dynamically evaluates the number of splits required. Constraint sets (2b) and (2c) are capacity constraints that enable the use of teaching and lab spaces beyond reduced capacity λ_ℓ. Constraint sets (2d) and (2e) regulate, in the event of a split, the number of students in such a split. Constraint sets (2f) and (2g) allow bottleneck course sections to be split and have capacity expanded, respectively. Constraint sets (2h) and (2i) are location and instructor conflicts for overlapping periods. Variable domains are expressed in (2k)–(2m).

3.4. Multiobjective framework

We propose four alternative objective functions for the DSCE formulation to prioritize efficient use of classrooms and honor instructor preferences. The primary objective function aims to maximize the number of course sections assigned, followed by minimizing the amount of capacity relaxation and lastly, minimizing the number of splits. The

values of ρ_1 and ρ_2 are intentionally set to hierarchically order the objectives, and are further discussed in Appendix.

$$\max\ f_1(\cdot) = \sum_{s\in S} w_s - \rho_1 \sum_{s\in S}\sum_{\ell\in\mathcal{L}_s} \theta_s^{\ell} - \rho_2 \sum_{s\in S}\sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s} z_s^{p\ell}. \tag{3}$$

The second objective function aims to maximize the (soft) preferences of instructors:

$$\max\ f_2(\cdot) = \sum_{s\in S}\sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s} v_s^{p\ell}(x_s^{p\ell} + z_s^{p\ell}). \tag{4}$$

Third, we aim to improve locational efficiency by minimizing the total number of teaching and lab spaces used, thereby fitting more course sections into the available space:

$$\min\ f_3(\cdot) = \sum_{\ell\in\mathcal{L}} y_\ell. \tag{5}$$

Our fourth objective, linearizable through a straightforward transformation, optimizes assignments in a manner that minimizes unused capacity in each location, seeking efficient utilization of all available resources:

$$\min f_4(\cdot) = \sum_{s\in S}\sum_{p\in\mathcal{P}_s}\sum_{\ell\in\mathcal{L}_s} \max\left\{0, \lambda_\ell c_\ell \left(x_s^{p\ell} + z_s^{p\ell}\right) - \hat{r}_s^n \left(x_s^{p\ell} + q_s^{p\ell}\right)\right\}. \tag{6}$$

The need to hierarchically manage multiple competing objectives and constraints contributes to the complexity of our framework in generating an optimized timetable. Among the various methodologies available to handle multiple objectives, we adapt a hierarchical method to prioritize interpretability [38]. The following multiobjective framework sequentially optimizes these objectives in any order of preference specified by decision makers:

$$\underset{x\in\mathcal{X}}{\text{minimize}} \quad f_i(x) \tag{7}$$

$$\text{subject to} \quad f_j(x) \le (1+\delta_j) f_j(x_j^\star),\ i = 1,2,\dots,k,\ j = 1,2,\dots,i-1. \tag{8}$$

where $\mathcal{X}$ represents the entire feasible region, i represents the specified order of objective function, $x^\star$ represents an optimal solution, and $f_j(x_j^\star)$ represents the optimal value of the jth objective function from the jth iteration. Small positive coefficient δ_j ensures the right hand side of constraint set (8) relaxes $f_j(x_j^\star)$ by a multiplier larger than one.[2]

4. Computational experiments

We now present the experimental framework to evaluate our mathematical optimization formulations and assess growth strategies using real-world datasets. All experiments were run using Gurobi Optimization 10.0.0 [39] and Python 3.11.6 on a high-performance computing cluster, with up to 128 GB memory, under Ubuntu 20.04.6. Our experiments revealed a diverse range of runtimes among instances for a given optimality gap tolerance. We thus developed and implemented a dynamic solver termination rule activated when the optimality gap exceeds 5% and runtime is more than 1 hour. It operates *after* an incumbent solution is found: the solver terminates if a reduction of ϵ% in the optimality gap is not found over a given time window (in minutes) t. As the instances we addressed exhibited rapid improvement in feasible solution quality over a relatively short time window, we found that setting $\epsilon = 0.05$ and $t = 20$ for the model with objective function $f_2(\cdot)$, $\epsilon = 0.03$ and $t = 60$ for $f_3(\cdot)$ and $\epsilon = 0.01$ and $t = 75$ for $f_4(\cdot)$ worked well.

4.1. Real-world datasets

We consider three distinct sets of data from Worcester Polytechnic Institute (WPI) for our case study, enabling the construction of test instances: *course registration* data, *student enrollment* data, and *distribution of major enrollment* data. The *course registration data* from Fall 2023 and Spring 2024 are used to understand the historical course sections that were assigned to physical locations and meeting time patterns. These data include features for each course section, such as academic term, assigned location and meeting pattern, instructor, enrollment count, instructional format, and delivery mode. These datasets are referred to as Fall and Spring data for the rest of the study. The *student enrollment data* includes all undergraduate students, including transferred students, and courses taken by each student in each term. In total, 11,315 students from 39 majors between 2015 Fall to 2023 Spring were analyzed. These data serve two purposes: first, to analyze and estimate each course section demand for each major and class, and second, to identify triplets of courses frequently taken together. For each major, class and term, a cross tabulation is generated that shows the frequency of following the corresponding path. Major changes, double majors, and minor programs are also considered. The *distribution of major enrollment data* details the population, such as first-year students, major program of study, and degree objective, such as Bachelor's of Science. We use this data to predict future major trends and conduct experiments on scaling university enrollments.

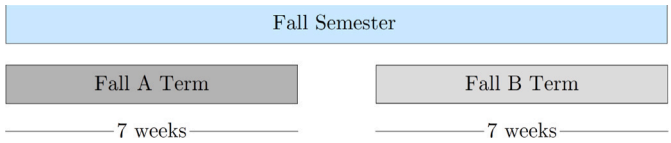


Fig. 1. Overlapping periods during Fall semester at WPI.

Table 4
Characteristics of course registration data used in our experiments.

Semester	$\lvert S\rvert$	$\sum_{s\in S} \hat{r}_s^{n=0}$	$\lvert\mathcal{P}\rvert$	$\lvert\mathcal{L}\rvert$	$\lvert\mathcal{I}\rvert$
Fall	1385	46,446	622	134	428
Spring	1304	40,551	616	134	442

At WPI, two overlapping periods exist in a single academic semester: one *term* over a seven-week duration and another *semester* over a fourteen-week duration occur simultaneously. The overlapping periods of the fall semester at WPI are shown in Fig. 1 (with a one week Fall break appearing between the 7-week terms).

4.2. Experimental design

We evaluate university expansion and contraction scenarios using the three datasets described in Section 4.1. Time slots for each day span from 8 AM to 9 PM in all instances. The characteristics of course registration data are provided in Table 4. When comparing the Fall data to Spring, there is a somewhat increased number of course sections and total enrollment.

Our novel framework enables universities to simulate various student body scenarios, accounting for shifts in major distributions and course trajectories over a specified time period. In our implementation, we adopt a four-year horizon, though this can be extended to any timeframe for planning purposes. We present our findings for the Fall 2027 and Spring 2028 semesters, which we refer to as *Fall+4* and *Spring+4*, respectively.

Estimating Major Distributions over Future Years. The choice of major largely determines the courses a student must complete, and changes in the distribution of majors can significantly affect scheduling decisions, particularly for courses with specific requirements or limited availability. To predict the major distribution of first-year students for each Fall semester from 2024 to 2027, we employed a Long Short-Term Memory (LSTM) model, a class of recurrent neural networks widely used in educational data mining for their ability to capture long-term dependencies, making them particularly effective for predicting student performance and dropout rates [40,41]. These forecasts enable us to accurately predict shifting demand for courses and classroom spaces.

[2] Any maximization objective $\{\max\ f(x)\}$ can be equivalently written as $\{\min\ -f(x)\}$, and the optimal objective function value afterward, negated.

Table 5
Characteristics of hard faculty preferences.

Reference	% of Course sections affected					$\lvert\mathcal{I}_{\text{Morning}}\rvert$					$\lvert\mathcal{I}_{\text{Middle}}\rvert$					$\lvert\mathcal{I}_{\text{Afternoon}}\rvert$				
	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4	0	1	2	3	4
Fall	0	7	15	25	35	0	10	23	48	120	0	20	46	50	50	0	10	23	48	57
Spring	0	7	15	25	35	0	9	21	45	90	0	18	42	55	55	0	9	21	45	65

The model is trained using the existing historical dataset on major enrollment distribution, which includes the total number of students enrolled in each major. This publicly available dataset covers the years 2014 to 2023 [42]. The LSTM model was selected for its superior accuracy in predicting 2023 data compared to other models such as linear and quadratic regression, exponential smoothing, and ARIMA. We assessed the accuracy of each model by summing the actual proportion of the student body represented by each major multiplied by the absolute error in the predicted number of students for that major. This metric was preferable over absolute deviation in the number of students, or the proportion of error, because absolute deviation prioritizes models that accurately predict large majors, while the proportion of error yields significant inaccuracies when small majors deviate by a small number of students. The LSTM predictions for 2024 to 2027 provide useful forecasts for the distribution of first-year students in each major and facilitate the estimation of future course demand based on enrollment levels.

Estimating Course Demand with Scaling Enrollment. In addition to major trends, another significant factor influencing course demand is the shift in academic paths. This effect is particularly prominent at WPI and other universities that emphasize flexibility in choosing unique trajectories according to student interests. We employed and configured exponential smoothing, a time series forecasting method known for its ability to effectively handle trends in historical data, with additive and damped trend components so as to forecast future registration ratios while maintaining computational efficiency. Our method accounts for the dynamic nature of student preferences and enrollment patterns. Regardless of how university enrollment rates fluctuate in future years, shifts in student preferences regarding majors and course selections necessitate adjustments in enrollment predictions. As the student population evolves, our approach reflects these changes by estimating the proportion of each student group (major and class standing) that enrolls in specific courses. This strategy allows us to generate enrollment estimates $\hat{r}_s^n$ for each course section s and change in first-year student body size n over the years 2024 to 2027.

Hard Faculty Preferences. Hard faculty preferences on daily availability, informed through direct consultations with university stakeholders, concern considerations such as child care and are categorized into five levels of severity across three time slots: 8 AM to 10 AM (morning), 10 AM to 1 PM (middle), and 1 PM to 5 PM (afternoon). As the severity level of hard preferences increases, increasing percentages of course sections and the number of instructors become unavailable, simulating potential disruptions in the schedule. No hard preferences are assumed for course locations. Details of hard faculty preference generation are given in Table 5.

Soft Faculty Preference Scores. Soft preferences reflect the desire of instructors to teach between certain time windows or preferred locations, which are not strict requirements but rather inclinations that are considered flexible. We simulated the soft preferences as follows. If a course section s is assigned to a preferred location ℓ and time pattern p, $v_s^{p\ell}$ takes a value of 1. If a course section is only assigned to the preferred location or time pattern, $v_s^{p\ell}$ is set to follow a uniform distribution, denoted as $v_s^{p\ell} \sim U[0,1]$, representing the degree of preference, whether simulated or elicited directly from instructors. However, if a course section is not assigned to a preferred location or time pattern, it does not add value to the total preference score, and $v_s^{p\ell}$ equals 0. This approach ensures that the scheduling process considers and prioritizes the soft preferences of instructors, aiming to create schedules that align with their preferences to the extent possible.

4.2.1. Experiments with baseline formulation

The baseline formulation is solved to provide insights into critical decision making questions. Given a set of courses, meeting patterns, preferences of instructors, course demands, and teaching and lab space capacities, the solution to the baseline formulation reveals which courses are assigned to which location and meeting pattern, and by extension which course sections become bottlenecks in the absence of certain buildings. We also highlight the effect of other factors such as reduced adjusted capacity, limited instructor availabilities, and forecasted major and course trends. The results reveal insights into the extent to which these factors lead to significant disruptions in the schedule.

Which Factor Has the Greatest Effect? An important question is the extent to which faculty preferences, adjusted capacity, and forecasted course demand affect university scheduling decisions in the subsequent academic years. We address this question by solving the baseline model (1) using the parameters specified in Table 6, considering five levels of faculty preferences shown in Table 5, six levels of capacity scaling factor, λ_ℓ, ranging from 70% to 120%, and scenarios both with forecasted and last year's course enrollments, then observing the resulting bottleneck sections.

Which Buildings Accommodate Near-Bottleneck Course Sections? Another important question decision makers may ask is the effect of closing certain buildings on the number of bottleneck sections, particularly in the context of long-term growth, increased enrollment, and changing trends. This analysis identifies which buildings are critical for accommodating near-bottleneck course sections and whether their closures, for example for renovation or repurposing, might lead to significant additional bottlenecks or disruptions in future timetables. To investigate this variation, we utilize the baseline model (1) for each expansion scenario across both semesters. We begin our simulations by setting the number of scaling students to $n = 0$, which we refer to as the *reference scenario*. We then examine the effect of building closures and increased the student population on the number of bottleneck sections

Table 6
Parameters for baseline experiments.

Symbol	Description	Values
α	Upper bound on the number of sections belonging to the same course that can be assigned during an overlapping time block	$\min_{c\in\mathcal{C}}\{(\lvert S_c \cap S_k\rvert \times 0.4), 10\}, \quad k = 1,2$
β	Upper bound on the number of sections belonging to the common course triplet g that can be assigned during an overlapping time block	$\lvert S_{(g,k)}\rvert \times 0.4, \quad k = 1,2$
γ	Upper bound on the number of sections of each course in the same instructional format that can be assigned during an overlapping time block	$\min_{c\in\mathcal{C},\, f\in\mathcal{F}}\{(\lvert S_c \cap S_k \cap S_f\rvert \times 0.4), 10\}, \quad k = 1,2$
ϵ	Optimality gap	0

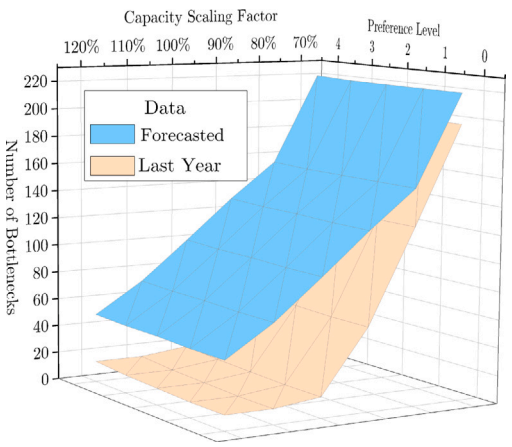


Fig. 2. Fall+4 semester. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

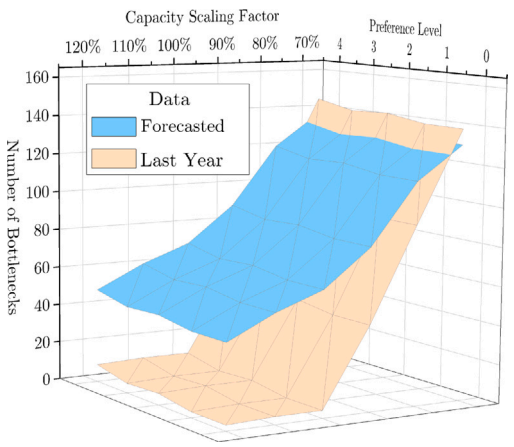


Fig. 3. Spring+4 semester. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

over the next four years, with $n = \{25, 50, 100, 125, 150\}$. We initially determine the popularity of buildings by examining the number of course sections assigned in the Fall and Spring schedules. Subsequently, we conduct simulations using the baseline formulation (1) with the parameters specified in Table 6 by independently and sequentially closing each building, one by one. For this experiment, we assume λ_ℓ to be 1 for both teaching and lab spaces, allowing full utilization of their capacities. We discuss the results of our experiments on the number of bottleneck sections for each enrollment growth scenario in the next section.

4.2.2. Experiments with DSCE formulation

The DSCE formulation is designed to address bottleneck sections in a twofold manner: by (1) relaxing the reduced capacity in teaching and lab spaces, and (2) splitting bottleneck sections into smaller sections that are assigned to suitable locations and time patterns. We conduct simulations using DSCE formulation (2), incorporating multiple objectives as outlined in Section 3.4, to model various contraction and expansion scenarios of the student body.

In the experiments, the percent of absolute physical capacity, λ_ℓ, is fixed at 70% for teaching spaces and 90% for lab spaces, while the model dynamically determines the number of splits per course section. We set a lower bound of 10 students and an upper bound of 80 students per split, based on discussions with stakeholders. These bounds ensure that class splits remain effective for learning and fit within the constraints of available space.

The metrics used to evaluate the results include the reduction in the number of bottleneck sections and overall resource utilization, with the dynamic model revealing tradeoffs between minimizing bottlenecks and the potentially costly need to open new sections. This evaluation provides decision makers with valuable insights for optimizing the allocation of course sections and effectively utilizing resources. While splitting bottleneck sections may incur significant costs such as hiring additional instructors, relaxing capacity is a more cost-effective strategy, yet subject to the decision maker's discretion on the extent of expansion. Hence, we choose the values of ρ_1 and ρ_2 in the objective function $f_1(\cdot)$ so that the model hierarchically optimizes in the following order: (i) maximize the number of course sections assigned, (ii) minimize the total capacity relaxation, (iii) minimize the number of splits. The derivations of ρ_1 and ρ_2 appear in Appendix. This approach allows decision makers to assess the balance between minimizing the number of bottleneck sections and managing the associated costs of splitting course sections. All experiments are run with the parameters given in Table 7.

5. Experimental results

We now discuss the results of our computational experiments of the optimization framework to assess long-term university capacity planning for different scenarios.

5.1. Three-way factor analysis

We conducted a series of experiments by solving our baseline formulation to reveal the impact of forecasted course demand, faculty preferences, and the capacity scaling factor, on university scheduling decisions. Figs. 2 and 3 visualize the results for the Fall+4 and Spring+4 semesters.

Figs. 2 and 3 indicate a significant impact of forecasted course enrollments, with increased number of bottlenecks when forecasts are used. This effect is particularly pronounced at higher levels of the capacity scaling, where the difference between the blue and orange surfaces is substantial. There is a clear inverse relationship between capacity scaling factor and the number of bottlenecks: as capacity decreases from 120% to 70%, bottlenecks increase.

The number of bottlenecks at corresponding capacity scaling factor is higher in the Fall semester than in the Spring semester. This difference is especially notable at lower capacities, where the Fall semester shows a steeper increase in bottlenecks. The impact of forecasted enrollments is also more pronounced in the Fall, indicating that accurate forecasting is even more critical during this semester to manage bottlenecks effectively in the long term.

The Spring semester shows a similar pattern to the Fall, with the inclusion of forecasted course enrollments significantly increasing the number of bottlenecks. However, at 70% adjusted capacity, the two surfaces intersect. This intersection suggests that at this level of capacity, the system reaches a point where the capacity constraint itself becomes the limiting factor. The primary limitation is insufficient instructional capacity, which leads to a similar number of bottlenecks regardless of whether forecasted enrollments are used. As capacity decreases from 120% to 70%, the number of bottlenecks increases consistently.

These experimental results indicate that both capacity scaling factor and forecasted enrollments are important factors influencing the number of bottlenecks in university scheduling. While faculty preferences do have an effect, it is subdued when compared to the other factors. This may be because there is a large enough number of instructors that, even under the most restricted preference setting, there is sufficient flexibility in locations and patterns for accommodating hard preferences. Collectively, these insights suggest that to improve scheduling efficiency, universities should prioritize accurate demand forecasting and consider strategies to enhance capacity flexibility.

Table 7
Parameters for DSCE experiments.

Symbol	Description	Values
U_s	Upper bound on the number of students in split s	80, $\forall s \in S$
L_s	Lower bound on the number of students in split s	10, $\forall s \in S$
μ_ℓ	Maximum allowed capacity expansion[a] for location ℓ	1.1, $\forall \ell \in \mathcal{L}$
n	Number of scaling students	−150, −100, −50, 0, 50, 100, 150
δ_1	Relaxion parameter of objective function $f_1(\cdot)$	0
δ_2	Relaxion parameter of objective function $f_2(\cdot)$	0.10
δ_3	Relaxion parameter of objective function $f_3(\cdot)$	0.20

[a] Additional seats can be added to expand beyond the actual capacity, though the extent varies by room size. This approach reflects adjustments observed during the COVID-19 pandemic, where classrooms were reconfigured to increase seating flexibility while maintaining safety.

5.2. Building closure analysis with baseline formulation

Certain buildings naturally possess greater teaching and lab space capacity, and the absence of any of such building will likely result in a greater number of bottleneck sections. As course sections may be scheduled in a limited number of locations, the effect of building absences on the resulting number of bottleneck sections can vary. Fig. 4 illustrates the number of bottleneck sections in Fall+4 and Spring+4 semesters, as buildings are independently closed for each expansion scenario under evolved major and course demands. Buildings appear on the x-axis in order of popularity for Fall and Spring semesters. Each bar indicates a particular expansion scenario under the absence of a single building specified on the x-axis. The dashed lines displayed on each bar represent the number of bottleneck sections in the reference scenario with all buildings open. The difference between the height of a bar and the dashed line represents the change in the number of bottleneck sections resulting from the absence of a specific building and shows the impact of the closure of that building.

In the Fall semester, Goddard Hall emerges as the most critical building across all scenarios, as its absence leads to a higher increase in the number of bottleneck sections. This is primarily because it accommodates specific laboratories such as chemistry labs to which some course sections must be assigned. On the other hand, the absence of some buildings such as Unity Hall and Kaven Hall has no additional effect on the number of bottlenecks when increased enrollment and shifting trends are considered.

Moreover, the figure shows an uneven rise in bottleneck sections across expansion scenarios between the two semesters. During the Spring semester, we observe a pattern where bottleneck sections tend to plateau and then increase sharply, in contrast to the more gradual distribution seen in the Fall as the student body grows Similar to the Fall semester, Goddard Hall remains a critical building. Its absence leads to a noticeable increase in bottleneck sections across all scenarios. In the Fall semester, we observe a balanced increase in the number of bottleneck sections across scenarios. The changes in bottleneck sections remain relatively consistent across different buildings, even with varying numbers of additional students. In contrast, the Spring semester demonstrates a more distinct pattern. The number of bottleneck sections increases steadily as 25 more students are added each year, up to a threshold of 125 additional students. Beyond this point, the increase in bottleneck sections becomes more pronounced, indicating a capacity threshold. This consistent rise in bottleneck sections across all buildings reflects a more uniform distribution of the impact of additional students in the Spring semester, up to the identified threshold. Buildings like Atwater Kent and Goddard Hall exhibit significant increases in bottleneck sections with higher numbers of additional students, suggesting potential capacity constraints that need to be addressed in case of possible closure. These observations highlight the distinct dynamics in the distribution of bottleneck sections between the two semesters. The Fall semester manages to maintain a balanced distribution of bottleneck sections despite varying student increases, while in the Spring semester, bottleneck sections steadily increase up to and including the 100 additional student expansion scenario. Beyond this point, the system struggles to accommodate further increases, resulting in a sharp rise in bottlenecks.

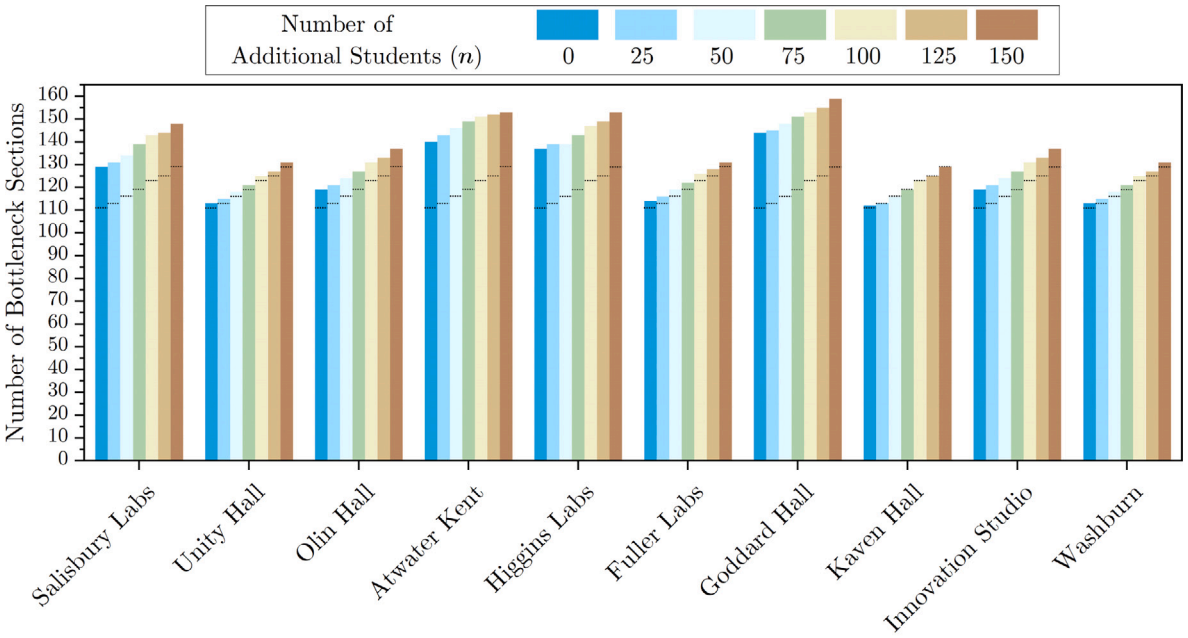
5.3. Addressing bottleneck sections via DSCE formulation

Table 8 shows the hierarchical optimization steps as outlined in the multiobjective framework of Section 3.4 using the DSCE formulation with a specific scenario from the Fall semester. We set the number of scaling students $n = 0$. This instance is selected solely for the purpose of illustrating *each* of the hierarchical steps involved in the formulation and does not imply generalizability or comparative analysis of the results. We begin by solving the baseline formulation (1) (Step **B**). This step is essential for generating parameter t_s for each course section $s \in S$, determining its ability to be split and whether the capacity of the assigned location can be expanded. As the hierarchical steps of the framework progress, namely steps **H1, H2, H3** and **H4**, the model prioritizes different objective metrics as bolded in Table 8. Step **H1** is the initial step of the hierarchical framework, which determines: (1) the number of additional course sections required through splitting, and (2) the total capacity violations. Notably, the number of capacity violations changes as the model progresses, while the total capacity violation, $\sum_{s\in S}\sum_{\ell\in\mathcal{L}_s}\theta_s^\ell$, remains fixed. The average violation, which indicates the extent to which the adjusted capacity exceeds 70% for teaching spaces and 90% for lab spaces, is adjusted accordingly. The final step **H4** reports metrics corresponding to the final schedule based on the chosen order of objectives.

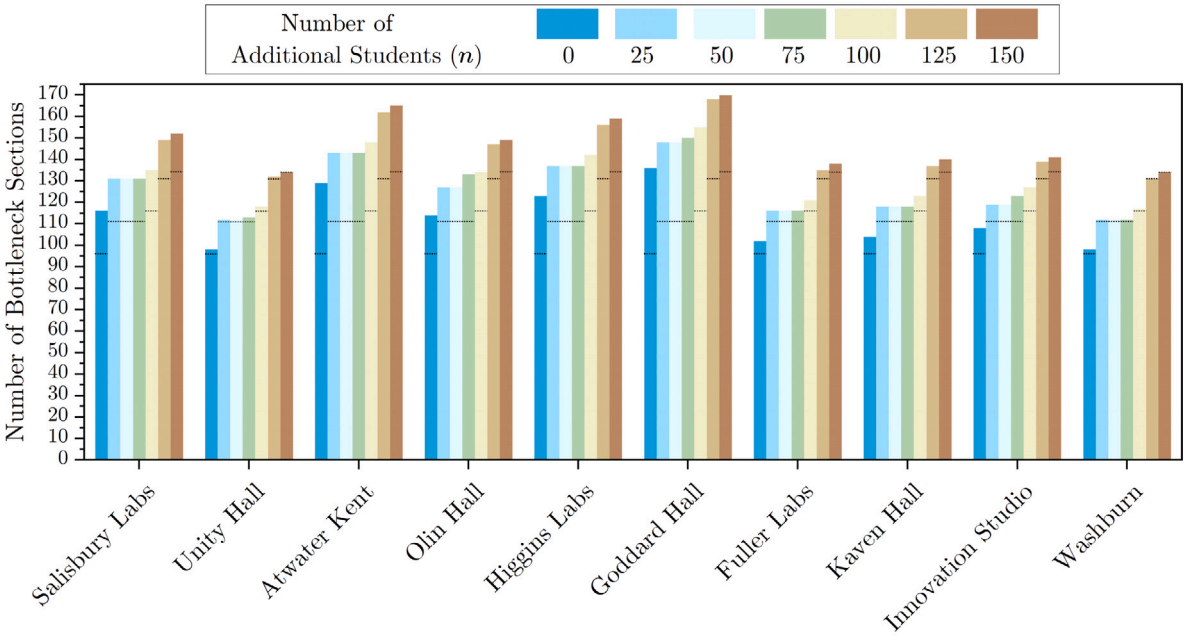
Step **H1** eliminates 75.9% of bottleneck sections by extending capacity beyond the adjusted limits in 60 courses and adding 57 course sections through splitting. In Step **H2**, where the total violation and number of additional course sections remain fixed, the model maximizes instructor soft preferences, achieving 81.5% of the maximum possible preference score. Step **H3** identifies the minimum number of locations by closing 36.8% of them, while meeting the relaxed total instructor preference score and accommodating the additional course sections. Finally, Step **H4** minimizes unused space in the utilized locations, ensuring efficient space use by reducing the number of seats in use by 58.9% and optimizing the distribution across more locations. Upon completion of Step 4, the DSCE formulation successfully addresses 75.9% of bottleneck sections through splitting and capacity expansion while utilizing 65.5% locations and 66.8% of available seats. Throughout all steps, the optimality gap and runtime remain reasonable for a strategic problem of this scale.

For other scenarios in Tables 9 and 10, we only report the baseline model results and the *final* step of the hierarchical dynamic optimization model to evaluate the effect of formulation (2) across different scaling scenarios. The runtime reported in Tables 9 and 10 in the final step is the summation of all hierarchical steps.

Tables 9 and 10 reveal that by employing objective (3), the dynamic formulation successfully eliminates 76.5% of the bottleneck sections from the baseline model in the Fall semester, and 81.5% in the Spring semester on average across all scenarios, achieved by allowing locations to be utilized beyond reduced capacity and, where needed, splitting course sections into available locations and time patterns. After allowing for capacity relaxation, our model also determines the minimally required number of additional course sections through splitting and the corresponding proportion of students to be assigned to each split.



(a) Fall+4 Semester.



(b) Spring+4 Semester.

Fig. 4. Effect of individual building closures on the number of bottleneck sections under various growth scenarios (as colored in key). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 8
DSCE formulation performance across hierarchical steps for Fall semester ($n = 0$).

Step	Number of Bottleneck Sections (Reference Scenario)	Number of Bottleneck Sections (Forecasted Data)	Number of Capacity Violations / Average Violation	Number of Additional Course Sections	Preference Score	Utilized Locations		Unutilized Locations		Runtime (sec)	Optimality Gap (%)
						Number of Locations	Number of Seats	Number of Locations	Number of Seats		
B	79	**141**	–	–	132/681	95	19,421	39	3,465	41	0
H1	21	**34**	**60/+12.1%**	**57**	179/681	97	17,408	37	3,450	922	0
H2	21	34	60/+12.1%	57	**555/681**	95	18,123	39	3,497	66	2.5
H3	21	34	62/+11.7%	57	499/681	**60**	21,673	74	4,791	462	5
H4	21	34	66/+11%	57	499/681	72	**8,906**	62	4,432	1,327	3.8

Table 9
DSCE formulation results for Fall+4.

Step	Scaling Enrollment (n)	Number of Bottleneck Sections (Forecasted Data)	Number of Capacity Violations / Average Violation	Number of Additional Course Sections	Preference Score[a]	Utilized Locations		Unutilized Locations		Runtime (sec)	Optimality Gap (%)
						Number of Locations	Number of Seats	Number of Locations	Number of Seats		
B	−150	134	–	–	121	96	19,676	38	3,455	39	0
H4		33	74/+8.3%	56	493	72	10,104	62	4,494	16,518	1.5
B	−100	138	–	–	133	93	20,967	41	3,532	33	0
H4		29	75/+9.5%	63	487	73	10,070	61	4,427	19,553	3.8
B	−50	138	–	–	125	96	19,596	38	3,462	38	0
H4		28	70/+11.1%	51	486	70	8,880	64	4,555	6,665	3.5
B	0	141	–	–	132	95	19,421	39	3,465	41	0
H4		34	66/+11%	57	499	72	8,906	62	4,432	2,820	3.8
B	50	146	–	–	133	95	18,467	39	3,553	39	0
H4		35	62 /+11.2%	60	786	72	8,663	62	4,421	5,368	4.5
B	100	154	–	–	121	95	20,169	39	3,496	41	0
H4		39	64/+10.6%	81	476	74	9,798	60	4,282	30,970	2.5
B	150	158	–	–	113	94	18,316	40	3,477	38	0
H4		40	63/+12.9%	85	485	73	9,715	61	4,349	22,148	7.1

Table 10
DSCE formulation results for Spring+4.

Step	Scaling Enrollment (n)	Number of Bottleneck Sections (Forecasted Data)	Number of Capacity Violations / Average Violation	Number of Additional Course Sections	Preference Score[a]	Utilized Locations		Unutilized Locations		Runtime (sec)	Optimality Gap (%)
						Number of Locations	Number of Seats	Number of Locations	Number of Seats		
B	−150	118	–	–	139	95	21,820	39	3,465	35	0
H4		21	49/+9.3%	48	502	69	11,303	65	4,685	2,663	4.7
B	−100	125	–	–	137	96	22,479	38	3,460	33	0
H4		25	52/+9.2%	49	497	69	10,650	65	4,677	2,069	4.3
B	−50	132	–	–	133	95	21,761	39	3,465	34	0
H4		25	61/+9.2%	46	492	68	9,876	66	4,661	1,718	2.6
B	0	141	–	–	137	95	21,771	39	3,495	32	0
H4		28	65/+10.5%	51	493	67	10,093	67	4,700	2,327	3.5
B	50	146	–	–	129	95	21,937	39	3,465	35	0
H4		28	66/+11.3%	53	490	69	9,588	65	4,669	2,155	2.5
B	100	162	–	–	141	6	21,047	38	3,460	33	0
H4		26	73/+10.1%	72	489	69	10,213	65	4,650	5,910	3.4
B	150	171	–	–	140	95	21,393	39	3,465	31	0
H4		30	79/+10.9%	75	484	68	9,890	66	4,679	2,188	3.9

[a] The maximum possible preference score is 681 and 670 respectively for all Fall and Spring instances.

The results demonstrate that when enrollment is scaled by n students over the next four years, the corresponding optimized assignments reveal the instructional space capacity of the university. Notably, additional course sections must be accommodated, even in contraction scenarios, to adapt to changing trends and to meet operational constraints. This offers the opportunity for delayed maintenance and renovation of unused locations that would otherwise remain idle for an entire semester. Even under the scenario with the highest projected growth, a substantial amount of instructional space remains underutilized in both semesters, and with this knowledge calling for intentional forethought into purposeful deployment.

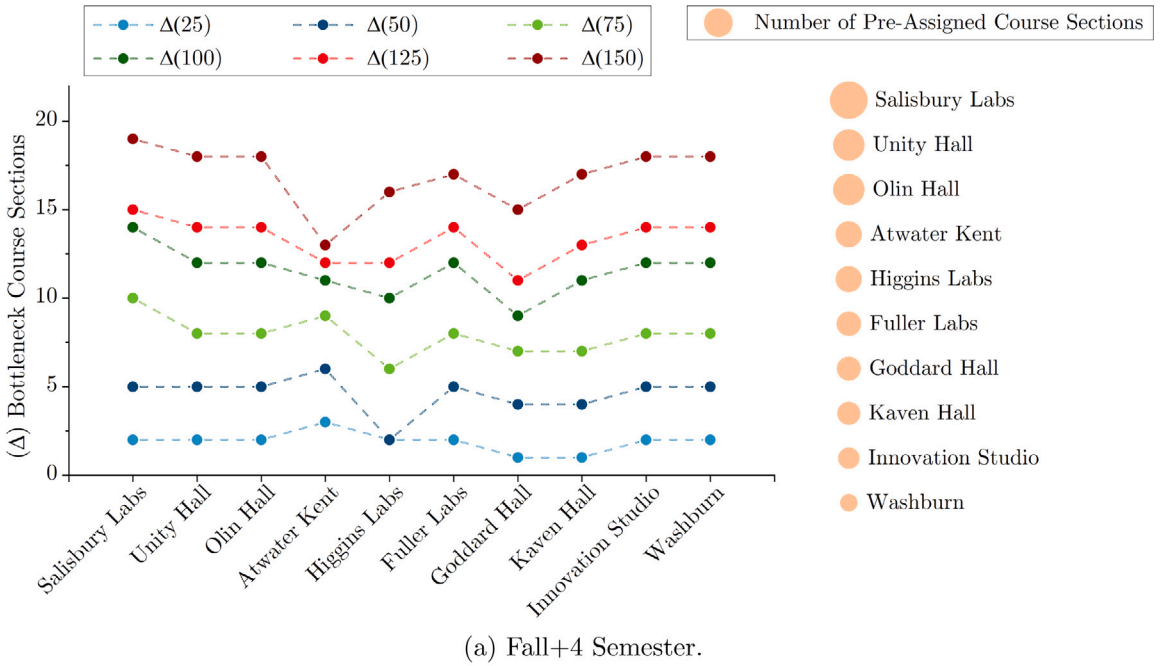
Through our optimization framework, we have identified that this underutilized space holds significant latent value, presenting a strategic opportunity for the university. Such space can be repurposed to address various needs, such as merging adjacent spaces to create adequate instructional areas, developing dedicated student study zones, or temporarily renting or loaning the space to generate revenue. This adaptability is made possible by leveraging our framework to maximize the utility of the campus infrastructure.

5.4. Case study implications

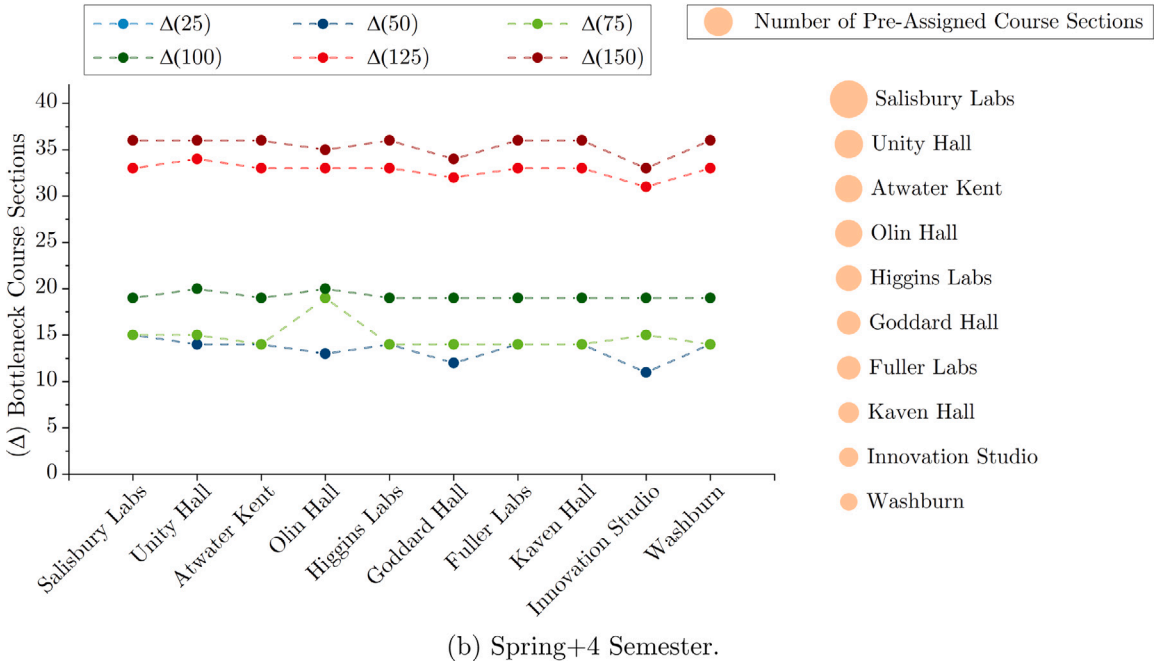
We now discuss key implications obtained from our experimentation. First, we discuss the marginal effect on bottleneck sections when more students are added in scenarios where key buildings are not available. Second, in the context of scaling enrollments, we highlight the tradeoff between the decrease in bottleneck sections and additional course sections. Next, we discuss how space utilization would have retrospectively improved under our optimization framework. Last, we provide insights for required splits and expanded capacity at the department level.

Long-Term Effect of Building Absences. Fig. 4 provides insight into the effect of building scarcity on bottleneck sections in each semester. We also evaluate the magnitude of changes between adjacent scenarios, measured by calculating the difference or delta (Δ) in the number of bottleneck sections between scenarios. Fig. 5 reveals the impact of adding n students in the individual absence of the ten most popular buildings in both Fall and Spring semesters. The size of each building circle in the right panel is proportional to the number of pre-assigned course sections in a reflection of their popularity.

Fig. 5 shows distinct patterns in the sensitivity of different building closures to additional students, with notable differences between the



(a) Fall+4 Semester.



(b) Spring+4 Semester.

Fig. 5. Marginal effect on number of bottleneck sections from adding students under absence of popular buildings. The size of the circles represents the relative popularity of the buildings.

two semesters. In the Fall semester, the sensitivity varies across buildings. For instance, Atwater Kent Laboratories shows minimal sensitivity to an increase from $\Delta(100)$ to $\Delta(150)$ students, indicating flexibility in accommodating higher enrollments in the event of closure. In contrast, other buildings display more notable changes in the number of bottleneck sections when faced with similar increments, reflecting greater sensitivity. In Spring, a different trend emerges. Most buildings show little to no sensitivity to smaller increments of 25 and 50 students, suggesting stable accommodation capacity for modest increases in the university body. However, a significant shift is observed between $\Delta(100)$ and $\Delta(125)$ scenarios, indicating a threshold beyond which reveals the physical capacity of the university is substantially challenged. When contrasted with the Fall semester, this discrepancy highlights the importance of considering semester-specific dynamics when planning for student body growth and building usage.

Tradeoff Between Bottleneck Reduction and Additional Course Sections. The reduction in the number of bottleneck sections often comes with the tradeoff of adding more course sections to the schedule through the process of splitting. However, the number of additional course sections does not increase at the same rate as the number of additional students n increases. A comprehensive analysis is necessary to assess, as n increases, the relationship between the reduction in the number of bottleneck sections and the associated increase in the number of required course sections.

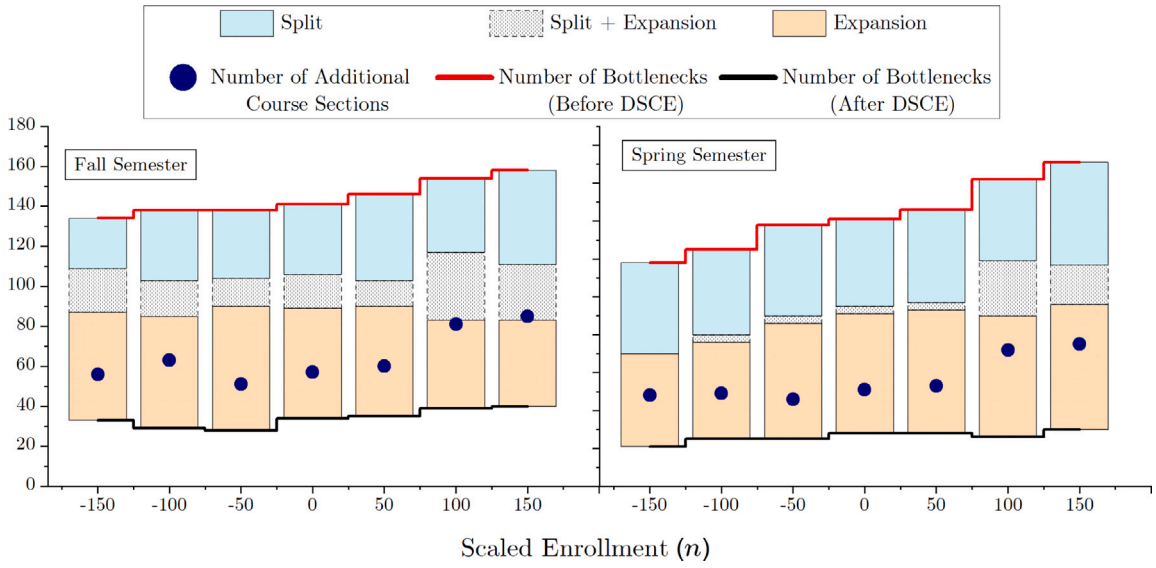


Fig. 6. The tradeoff, as the number of students n increases, between the number of additional course sections required and bottleneck sections as observed in Fall (left) and Spring (right) semesters. The number of splits, capacity expansions, and split + capacity expansions are indicated.

Fig. 6 shows the dynamic model performance in addressing bottlenecks via splitting and relaxing capacity in Fall+4 and Spring+4 semesters. It illustrates how bottleneck sections are resolved through splitting, capacity expansion, or both. In each of the Fall and Spring semesters, even during contraction, splitting is needed to address bottleneck sections. The number of additional course sections does not always increase with the number of students. This is primarily because the number of bottlenecks is determined by the baseline model for each scenario, and these bottlenecks may exhibit different splitting behaviors across different scenarios. For example, one course may be split into four sections, while another course may not be split at all due to the limited availability of the instructor. Upon the addition of 100 students, a substantial increase in the number of additional course sections, as well as in the hybrid solution where both splitting and expansion are applied, is observed in both semesters. This indicates that the dynamic model effectively addresses the majority of bottleneck sections without a significant increase in the number of splits, at least up to a certain threshold.

These observations highlight that universities can carefully weigh strategic increases in student enrollment while implementing relaxed capacity and dynamic splitting procedures to effectively prioritize resource utilization, minimize scheduling conflicts, and enhance the overall efficiency of the scheduling process.

Actual versus Optimized Space Utilization. Our framework effectively addresses the challenge of accommodating scaling enrollment over four academic years while ensuring the efficient allocation of university resources. To further evaluate other potential benefits of our dynamic formulation, for the purposes of this experiment we exclude predicted course demand, reduced capacity and hold enrollment numbers constant to conduct a comparative analysis between the actual and optimal assignments obtained under the reference scenario.

Fig. 7 provides a visual representation of the utilization and occupancy levels of each instructional space in a typical week in Fall semester, A term. The x-axis represents the total room occupancy in hours, while the y-axis depicts the percent utilization. The diameter of each data point corresponds to the capacity of the room. In the actual schedule, a total of 92 instructional spaces were in use. These spaces had an average occupancy of 20.6 h per week, with an average utilization rate of 65%, as indicated by the red dashed lines. Notably, certain laboratory and teaching spaces were utilized well beyond their 100% capacity, contributing to the overall average utilization value. This is achieved by adding extra seats to locations when necessary. In contrast, the optimized resource assignments from our dynamic formulation lead to a substantial improvement in resource allocation while using only 75 locations. The average occupancy of rooms in a typical week is increased to 25.8 h, reflecting a significant 25.2% improvement. Our analysis also reveals a more balanced allocation of resources, with utilization rates primarily influenced by our capacity adjustment multiplier set at 100% for both lecture and lab course sections. This adjustment results in 18.46% improved utilization rates distributed around a new mean of 77%, with a tighter clustering of instructional spaces around the mean. When applied to the DSCE formulation, our framework results in a more efficient use of space, with less variability and fewer instances of extreme under- or over-utilization compared to the actual scenario. Notably, the upper-right corner can be seen as the optimal area, where utilization exceeds 80% and total occupancy is more than 25 h per room. DSCE outperforms the actual assignment, with 33% of locations falling within this region, compared to only 2% in the actual assignment.

Fig. 8 provides an alternative view of the distribution of utilization across the campus for both Fall A and Spring C semesters. The x-axis represents the percentage of space utilization, while the y-axis shows the count of instructional spaces for every course-location assignment. We observe that the optimization significantly increases the number of spaces utilized at close to full capacity and reduces the occurrences of underutilized or overutilized spaces, reflecting a more efficient allocation of resources.

Implications for Strategic University Planning. Long-term planning decisions, such as hiring new faculty members and expanding lab and lecture spaces, are a collaborative process between upper-level administration and individual departments. Departments express their projected needs based on enrollment trends and academic priorities, while the administration allocates budgeted resources to meet those needs within institutional constraints. Consequently, we first analyze the number of splits required under evolving major distributions and student course trajectory preferences within each department. Next, we examine the types of spaces where capacity expansion is needed to address bottleneck sections. For both analyses, we consider two scenarios: (1) a reference scenario with $n = 0$, and (2) a growth scenario where $n = 150$.

Fig. 9 depicts total number of splits required in the Fall+4 semester. In both scenarios, the majority of bottlenecks are resolved by splitting

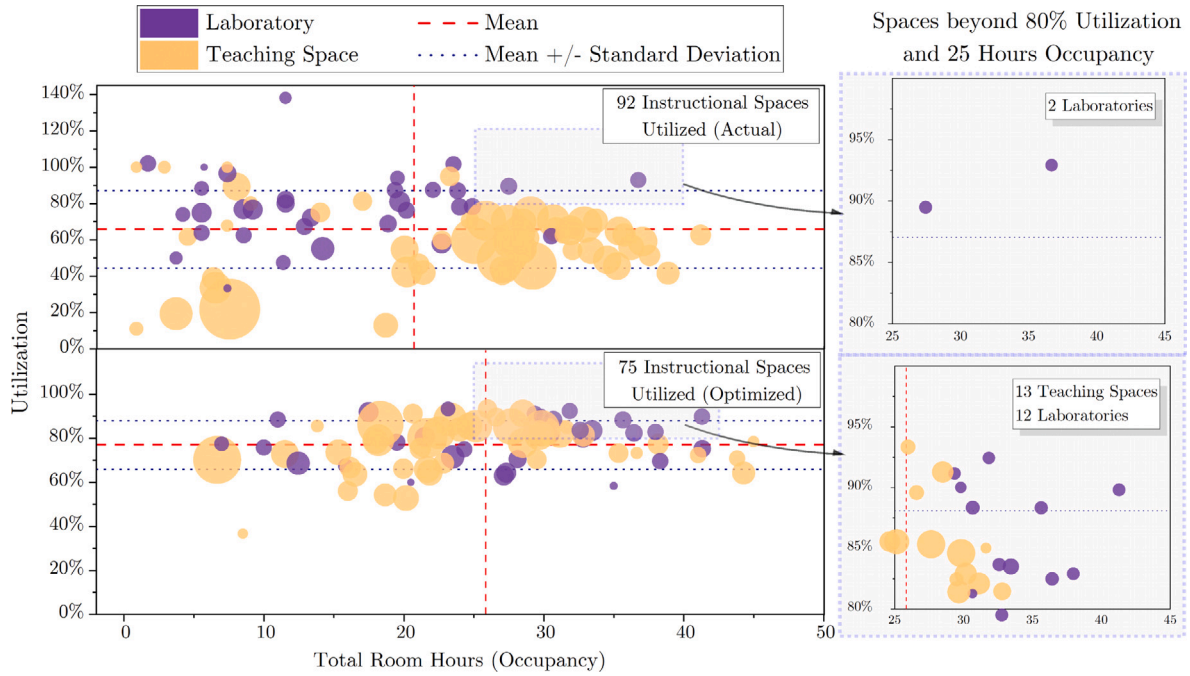


Fig. 7. Utilization and occupancy of instructional space during Fall semester, A term. The size of each circle is proportional to the capacity of the corresponding location. Red dashed lines show the average value of each axis. Blue dotted lines show the average value +/− standard deviation. The number of instructional spaces utilized is shown in the figure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

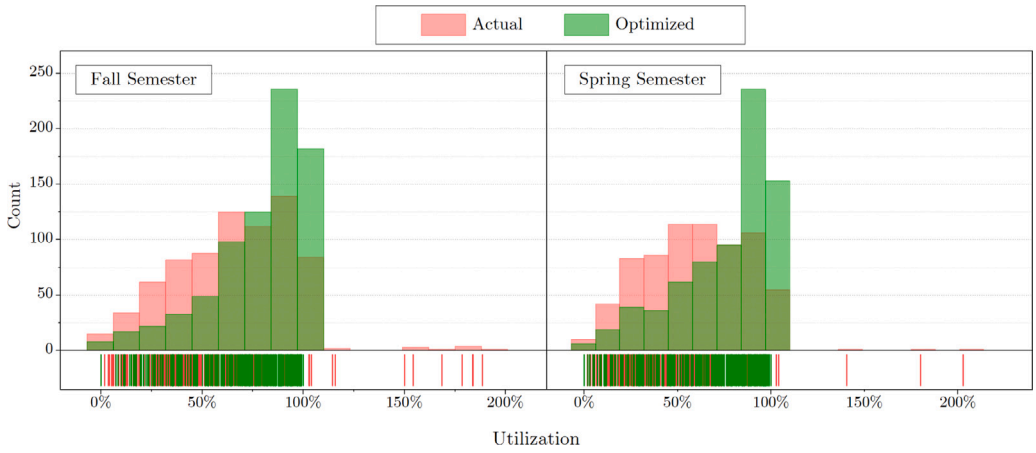


Fig. 8. Histogram of utilization of instructional spaces during Fall semester, A term (left) and Spring Semester, C term (right).

courses into two smaller sections. Even in the reference scenario, additional course sections are needed in Computer Science, Chemistry, and Biomedical Engineering due to shifted course trajectories and major distributions over the four-year period. This highlights the need to hire new faculty members if student preferences are to be accommodated. On the other hand, more splits are required in certain departments as the university adds 150 students annually over the four-year period. This observation allows universities to make informed decisions regarding student body growth.

Fig. 10 illustrates the splits across departments in the Spring+4 semester and highlights the differences between the two semesters. In the Spring semester, fewer departments are required to accommodate additional course sections. This disparity may result in an uneven distribution of faculty course loads in the event of new hires.

Another important long-term investment is the decision to renovate unused spaces or construct new laboratories and lecture rooms. The Fall semester is the primary constraint due to higher enrollments and more restrictive space availability. Since Fall violations already account for most capacity issues, Spring requires fewer adjustments. Expanding capacity based on Spring violations alone would be ineffective, as Fall would still need expansions. Therefore, we focus on Fall to ensure a comprehensive solution. Fig. 11 illustrates the capacity expansion percentages across different types of instructional spaces in the reference scenario ($n = 0$) and growth scenario ($n = 150$) for the Fall+4 semester. The analysis reveals varying degrees of capacity expansion across instructional spaces in response to an increasing student population and shifted demand over the next four years. Electronics laboratories, in particular, emerge as a key area requiring investment due to their significant potential for capacity increases. Similarly, computer laboratories show notable growth, indicating the need for moderate expansion to accommodate higher demand. Interestingly, lecture rooms exhibit a contrasting trend. While their capacity expansions are prominent in the reference scenario, they decline significantly under the growth scenario. This may be attributed to large classes becoming bottlenecks as student numbers rise, resulting from instructor conflicts or capacity limitations. Therefore, future efforts should focus

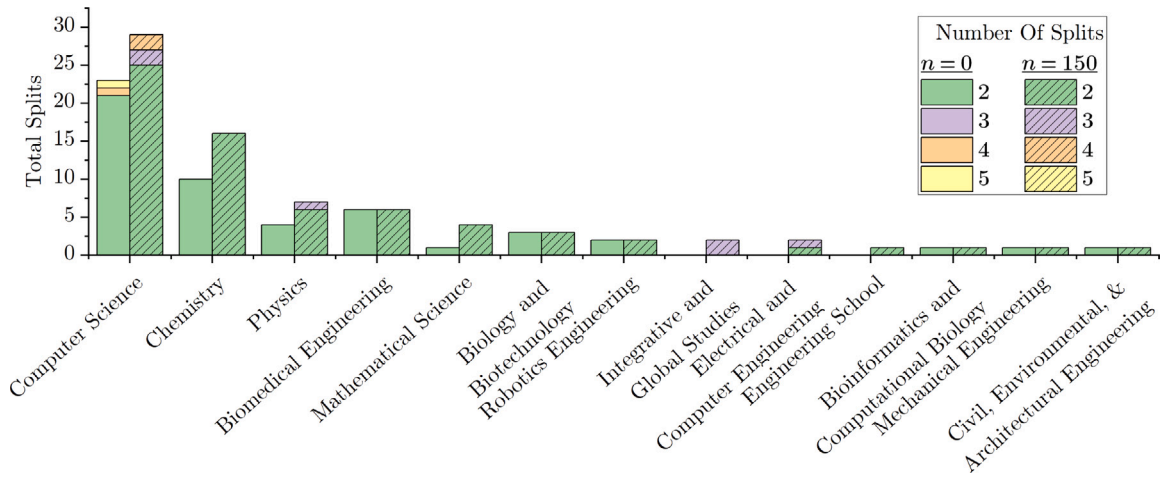


Fig. 9. Number of splits required for each department in reference scenario, $n = 0$, and growth scenario $n = 150$ in the Fall+4 semester.

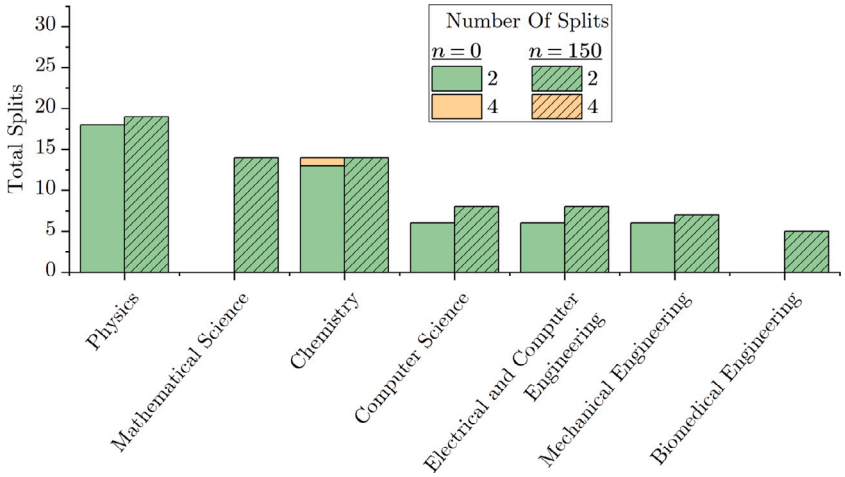


Fig. 10. Number of splits required for each department in reference scenario, $n = 0$, and growth scenario $n = 150$ in Spring+4 semester.

on renovating unused spaces or hiring additional faculty, rather than directly expanding lecture rooms. Biology laboratories display a moderate capacity expansion need in the reference scenario, highlighting the potential shifted demand for corresponding classes. Robotics laboratories demonstrate growing demand that may increase in the future. Finally, chemistry and physics laboratories show relatively lower capacity expansion needs; however, chemistry laboratories, in particular, may still require attention to prevent future overcrowding.

6. Conclusion

Our study is the first of which we are aware to optimize university-wide course scheduling in light of fluctuating student enrollments, shifting major distributions, and evolving course trajectories, aiming to improve instructional space management and student body planning in the long term. We develop a framework based upon multiobjective mathematical optimization that aims to balance various competing factors such as space utilization and instructor scheduling preferences. Central to our framework is the treatment and resolution of *bottleneck* course sections, critical for optimizing resource allocation in universities. We develop an extended dynamic formulation that addresses bottleneck sections through variables that allow for expanding available capacity as well as splitting bottleneck sections into smaller sections, or *splits*.

Our formulation employs multiple objectives including minimizing bottleneck sections, split course sections, and capacity expansion, while maximizing soft instructor preferences and minimizing unused capacity and the number of locations. It integrates real-world constraints such as common course section conflicts and instructor time window availability, that are often omitted in the literature due to computational tractability. Able to be adopted by different institutional contexts and scenarios, our framework allows decision makers to adjust parameters such as the priority of objectives, the minimum number of students in splits, capacity scaling, and building closures. Our framework can easily adapt to different enrollment scenarios that enable university decision makers to fine tune strategies for planning ahead, regardless of student body growth or contraction. We further provide insights into tradeoffs between fluctuating student body size and required additional course sections, supporting dynamic data-driven decisions that balance instructional space needs with broader campus infrastructure requirements.

The optimization framework offers several benefits to the university, contributing to long-term sustainability, resource savings such as reduced operational costs and a reduced environmental footprint, and stakeholder well-being. By optimizing instructional space, the framework reduces the need for additional infrastructure, leading to potentially lower operational costs and a reduced environmental footprint. Additionally, the framework enhances the well-being of students and faculty by strategically recommending capacity expansions and accommodating necessary course splits, which prevents overcrowding and last-minute schedule disruptions. By reducing bottlenecks and minimizing scheduling conflicts, instructors experience more predictable

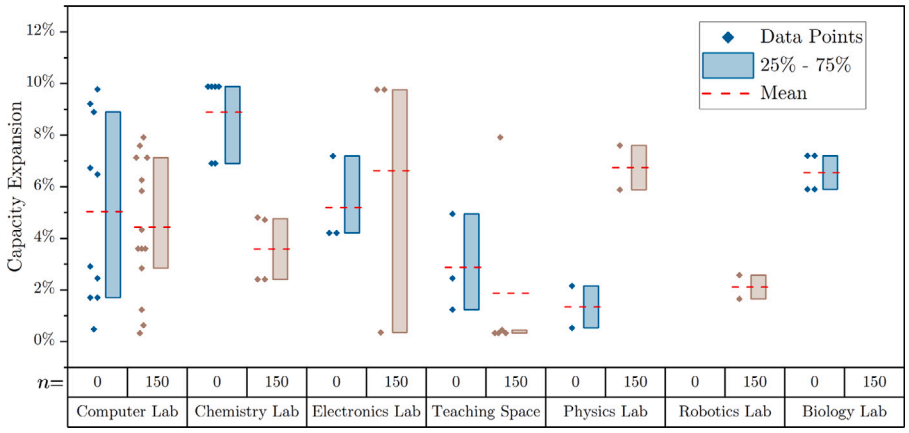


Fig. 11. Capacity expansion percentages across different types of instructional spaces in reference scenario, $n = 0$, and growth scenario $n = 150$ in Fall+4 semester.

teaching schedules, and students gain access to better course availability and improved learning environments. These improvements create a more efficient, adaptable, and sustainable university scheduling system that benefits all stakeholders.

With the advancements our framework achieves, come some limitations. Though our dynamic formulation often eliminates many bottleneck sections, there may still be sections that cannot be assigned to any suitable time pattern and location due to various constraints such as limited capacity, scheduling conflicts, or hard faculty preferences. As our results indicate that otherwise unavoidable hard faculty preference conflicts are relatively few, we leave them to be resolved the way they always have been, that is, through collegial discussion between parties involved in the resolution. This may include further exploration of unused locations and seats, or potential transformation of unused locations into required spaces. For any institution intent on carrying out this work in practice, an important consideration is the intentional elicitation of instructor preferences, both hard and soft, to inform the framework. This can be achieved by the deployment of a user-friendly platform that can include a number of possible expansions to this study. Finally, the scope of our study was on the use of academic spaces to schedule courses. While we realize that other primary resources can constrain growth such as dormitories, dining facilities, and personnel, we simply maintain that any expansion in student body size should be accompanied by commensurate investments in resources, including faculty, staff, student support, and necessary infrastructure.

In summary, we present a significant advancement in the area of university space management and in particular, the scheduling of courses into instructional spaces while considering scaling enrollments plus trends in major and course demands. Our multiobjective optimization framework provides a practical and adaptable solution to these challenges. By allowing for dynamic splitting and capacity expansion to handle bottleneck sections as well as space constraints and instructor preferences, our framework represents a realistic and customized approach for academic scheduling that remains eminently adaptable to other academic settings. Our study contributes to the field by addressing complex and university-wide timetabling, while offering insights into long-term, flexible planning for educational institutions. It is our hope that this framework serves as a valuable tool for universities, empowering informed, data-driven decisions in an ever-evolving educational landscape.

CRediT authorship contribution statement

Özge Aygül: Formal analysis, Methodology, Writing – original draft, Conceptualization, Visualization. **Teodor Hellgren:** Writing – original draft, Validation, Investigation, Formal analysis. **Shima Azizi:** Writing – review & editing, Supervision, Investigation. **Andrew C. Trapp:** Methodology, Investigation, Conceptualization, Writing – review & editing, Writing – original draft, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to proofread the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Lexicographic objective function coefficients

Formulation (3) represents a lexicographic objective function that hierarchically prioritizes three objective components. The values of ρ_1 and ρ_2 are carefully chosen to establish the desired ordering of priorities in a hierarchical manner. The primary objective aims to maximize the total number of course sections assigned, while the secondary objective is concerned with relaxing the reduced location capacity for splittable course sections. The third objective aims to find the minimum number of splits necessary to address bottleneck sections. For the purpose of deriving coefficients, we observe the number of bottleneck sections, denoted as $|\hat{S}|$, from solving baseline formulation (1) as well as the lower and upper bounds on the number of splits, $\underline{L}_s$ and $\overline{U}_s$ respectively.

First, we require that the maximum contribution to the objective function of capacity relaxation be strictly less than the minimum contribution of splitting any course section to ensure relaxing capacity is preferable to splitting any section: $\rho_1 \sum_{s\in\hat{S}} \sum_{\ell\in\mathcal{L}} \theta_s^l < \rho_2 \sum_{s\in\hat{S}} \sum_{p\in\mathcal{P}_s} \sum_{\ell\in\mathcal{L}_s} z_s^{p\ell}$.

Second, we aim to guarantee that the maximum contribution of relaxing reduced capacity and splitting course sections together is strictly less than the minimum contribution of assigning a course

section: $\rho_1 \sum_{s\in\hat{S}} \sum_{\ell\in\mathcal{L}} \theta_s^l + \rho_2 \sum_{s\in\hat{S}} \sum_{p\in\mathcal{P}_s} \sum_{\ell\in\mathcal{L}_s} z_s^{p\ell} . < 1$. To do so, it is important to determine the ρ_2 component in terms of ρ_1, followed by solving for the value of the ρ_1 component.

We can thereby obtain values of ρ_1 and ρ_2 by solving the following linear system:

$$\rho_1 \ |\hat{S}|(\mu_l - \lambda_\ell) + \epsilon - \rho_2 \ \underline{\mathcal{L}}_s = 0,$$
$$\rho_1 \ |\hat{S}|(\mu_l - \lambda_\ell) + \epsilon + \rho_2 \ |\hat{S}| \ \overline{\mathcal{U}}_s - 1 = 0,$$

where we set the value of ϵ to be a small positive constant. Solving for ρ_1 and ρ_2, we find:

$$\rho_1 = \frac{-\underline{\mathcal{L}}_s\epsilon + \underline{\mathcal{L}}_s - |\hat{S}|\overline{\mathcal{U}}_s\epsilon}{|\hat{S}|(\mu_\ell - \lambda_\ell)(\underline{\mathcal{L}}_s + |\hat{S}|\overline{\mathcal{U}}_s)},$$

$$\rho_2 = \frac{1}{\underline{\mathcal{L}}_s + |\hat{S}|\overline{\mathcal{U}}_s}.$$

Data availability

No data was used for the research described in the article.

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